

Course Summary

RECURRING THEMES

Methods for Finding Data

1.1	Linking in Hierarchical & Network data models
1.2	Pointers vs. Foreign keys
5.2	Relational DB indices
6.4	MongoDB indices
6.4	MongoDB design: standalone vs. embedded documents

Complex vs. Simple Design

2.6	ERD	vs.	Antipatterns
A4	Our design		Ancestry.com
4.2	Business logic in DB		In application
6.1	Inmon-style		Kimball-style data warehouse
6.2	Relational		OLAP, NoSQL

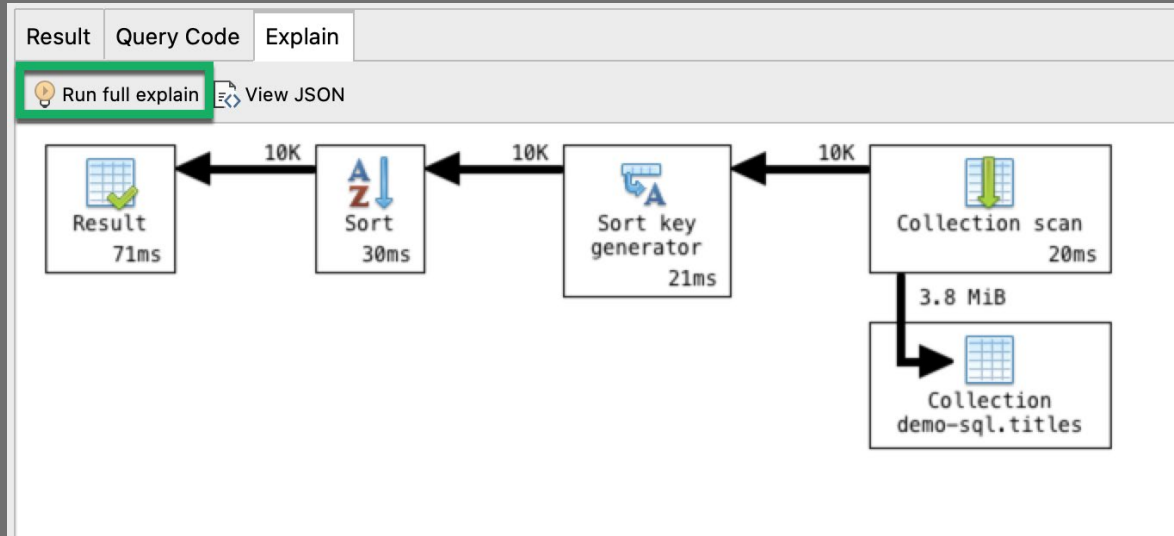
Data Integrity vs. Efficiency

A2	Our design	vs.	Ancestry.com
6.1	OLTP		OLAP
6.1	Normalize		Denormalize
6.2	ACID		BASE
6.4	ERD		Dimensional Modeling MongoDB Design

WHAT DID THIS COURSE COVER?

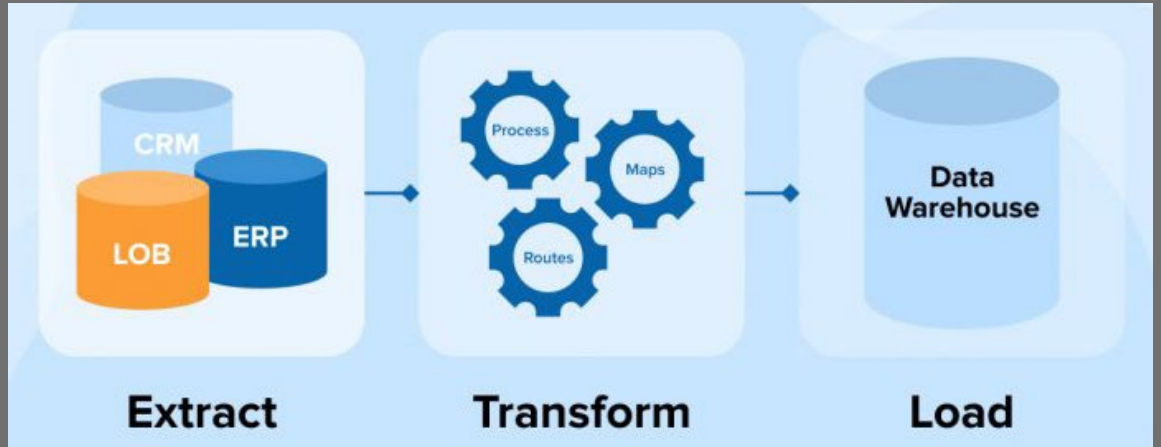
Database Theory

- Functions & relations
- Relational calculus & algebra
- Grouping
- Aggregation
- Query plans



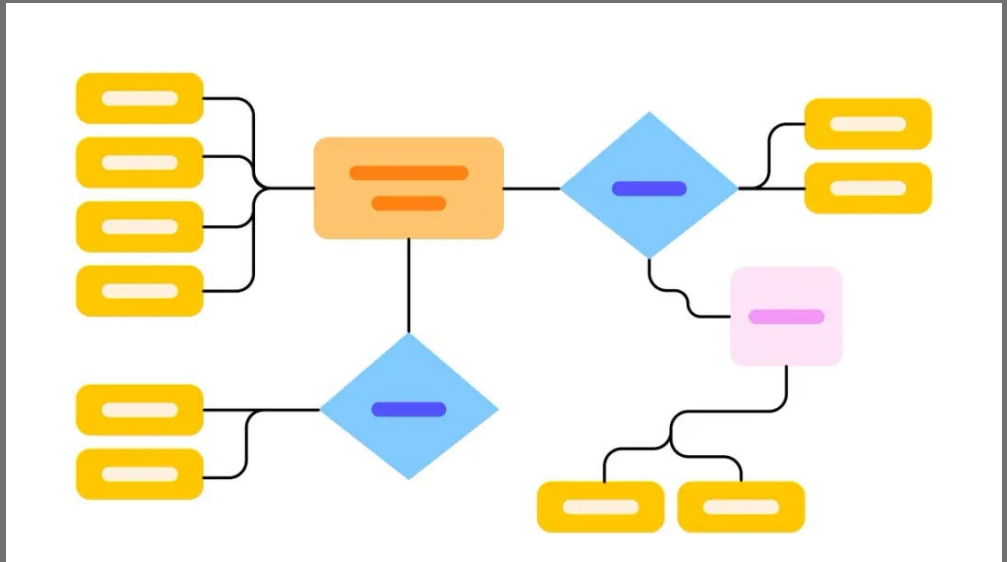
Database Programming

- SQL, PyMongo, Cypher
- Query costs & efficiency
- Database/application integration
- Security
- ETL



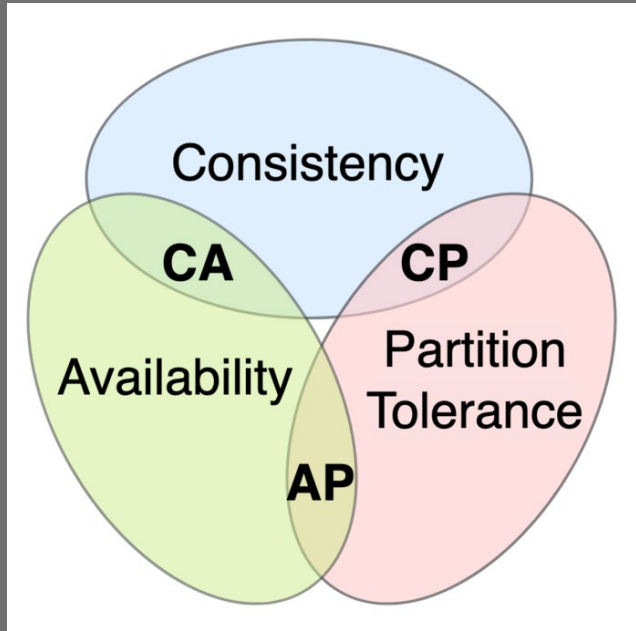
Database Design

- Design in OLTP, OLAP, MongoDB, Neo4J
- Normalization & denormalization



Concurrency in Databases

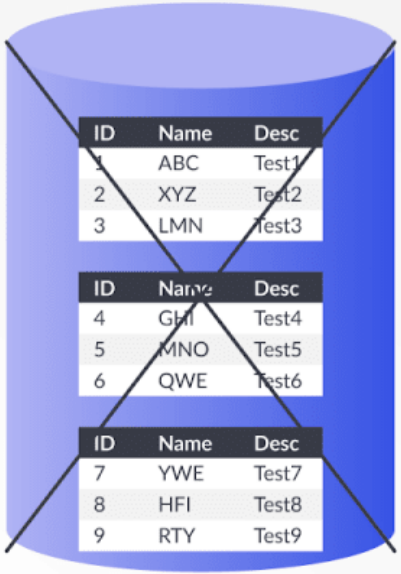
- Serialization
- Transactions & isolation levels
- ACID & BASE properties
- CAP theorem



BEYOND THIS COURSE

Big Databases

PARTITIONING



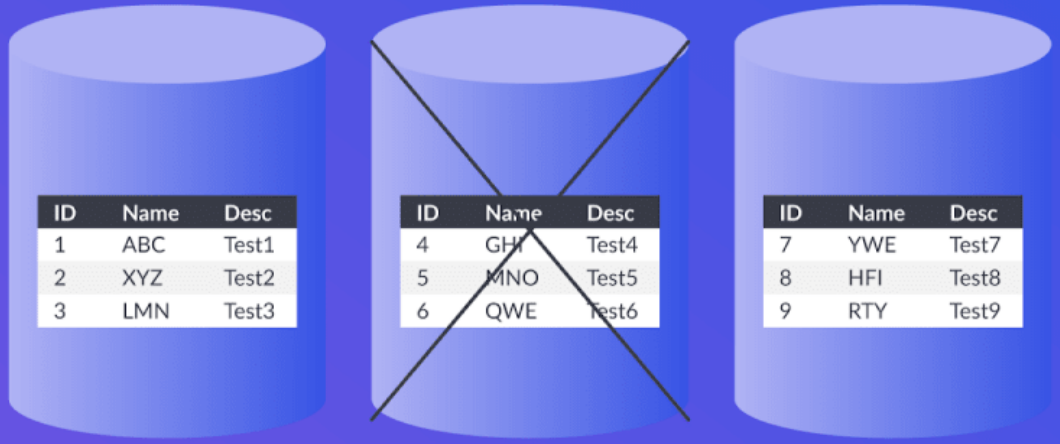
ID	Name	Desc
1	ABC	Test1
2	XYZ	Test2
3	LMN	Test3

ID	Name	Desc
4	GHI	Test4
5	MNO	Test5
6	QWE	Test6

ID	Name	Desc
7	YWE	Test7
8	HFI	Test8
9	RTY	Test9

Assume this machine goes down!
All partitions unavailable

SHARDING



ID	Name	Desc
1	ABC	Test1
2	XYZ	Test2
3	LMN	Test3

Shard1

Still available

ID	Name	Desc
4	GHI	Test4
5	MNO	Test5
6	QWE	Test6

Shard2

Assume this machine goes down

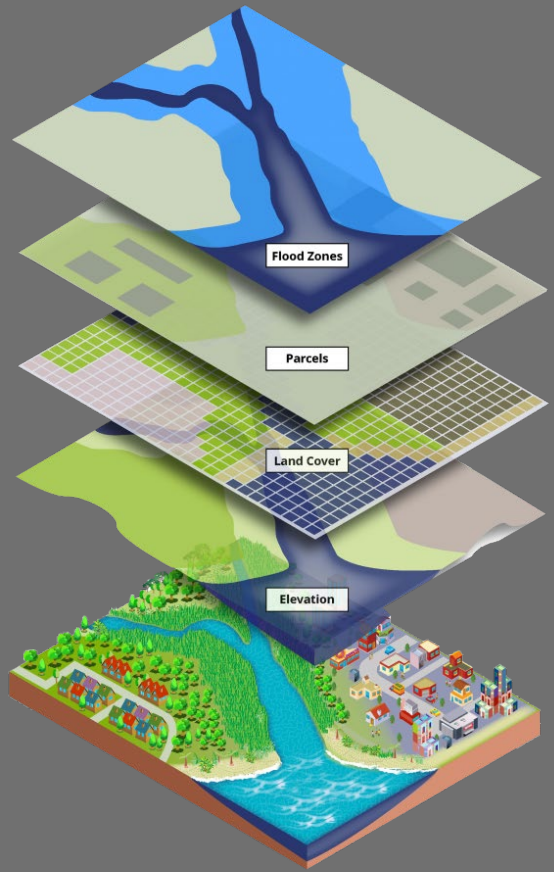
ID	Name	Desc
7	YWE	Test7
8	HFI	Test8
9	RTY	Test9

Shard3

Still available

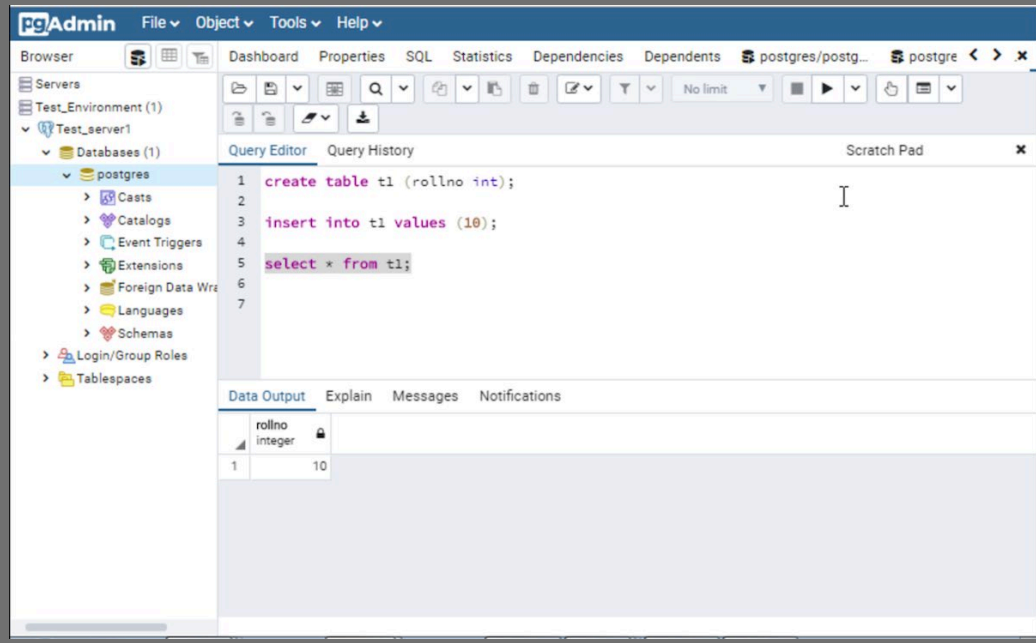
Other Database Models

- GIS
- Time-series
- BI (Business Intelligence)
- Key-value store
- Column store



Tools

- pgAdmin
- MongoDB Compass
- Neo4J Desktop



Practical Issues

- OLAP language MDX
- Live migration
- Realistic ETL
- Data warehouses
- Data lakes
- Cloud-based DBs

